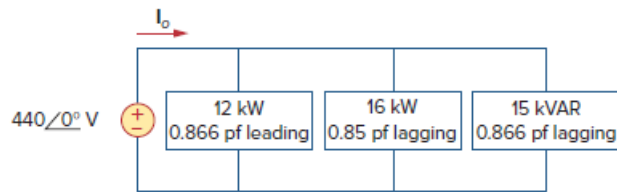


Problem

Find I_o in the circuit of Fig. 11.82.

Figure 11.82:



Step-by-step solution

Step 1 of 17

Refer to Figure 11.82 in the text book.

Consider the circuit diagram shown in Figure 1.

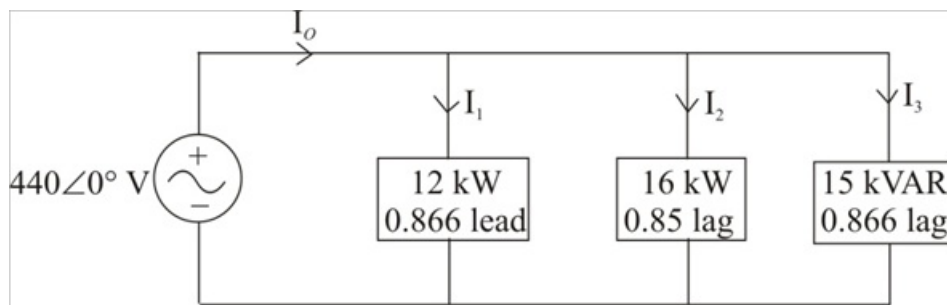


Figure 1

Step 2 of 17

Calculations for load-1:

Calculate the power factor angle at load 1.

$$\begin{aligned}\theta_1 &= \cos^{-1}(0.866) \\ &= 30^\circ\end{aligned}$$

Therefore, the value of the power factor angle θ_1 is 30° .

Step 3 of 17

Calculate the apparent power absorbed by load 1.

$$S_1 = \frac{P_1}{\cos \theta_1}$$

Substitute 12 kW for P_1 and 30° for θ_1 .

$$\begin{aligned} S_1 &= \frac{12 \text{ k}}{\cos 30^\circ} \\ &= \frac{12 \text{ k}}{0.866} \\ &= 13.857 \text{ kVA} \end{aligned}$$

Therefore, the value of the apparent power S_1 is, 13.857 kVA.

Step 4 of 17

Calculate the reactive power absorbed by the load 1.

$$Q_1 = -S_1 \sin \theta_1$$

Reactive power is negative since the power factor is leading.

Substitute 13.857 kVA for S_1 and 30° for θ_1 .

$$\begin{aligned} Q_1 &= -(13.857 \text{ k}) \sin(30^\circ) \\ &= -(13.857 \text{ k})(0.5) \\ &= -6.928 \text{ kVAR} \end{aligned}$$

Therefore, the value of the reactive power Q_1 is -6.928 kVAR.

Step 5 of 17

Calculate the complex power absorbed by the load 1.

$$S_1 = P_1 + jQ_1$$

Substitute -6.928 kVAR for Q_1 and 12 kW for P_1 .

$$S_1 = (12 \text{ k} - j6.928 \text{ k}) \text{ VA}$$

Therefore, the value of the complex power absorbed by the load 1 is,

$$(12 \text{ k} - j6.928 \text{ k}) \text{ VA}.$$

Step 6 of 17

Calculate the current passing through the load 1.

$$I_1 = \left(\frac{S_1}{V} \right)^*$$

Substitute $(12 \text{ k} - j6.928 \text{ k}) \text{ VA}$ for S_1 and $440 \angle 0^\circ \text{ V}$ for V .

$$\begin{aligned} I_1 &= \left(\frac{12 \text{ k} - j6.928 \text{ k}}{440 \angle 0^\circ} \right)^* \\ &= (27.273 - j15.746)^* \\ &= 27.273 + j15.746 \\ &= 31.492 \angle 30^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current flowing through load 1 is, $31.492 \angle 30^\circ \text{ A}$.

Step 7 of 17

Calculations for load-2:

Calculate the power factor angle at load 2.

$$\begin{aligned}\theta_2 &= \cos^{-1}(0.85) \\ &= 31.79^\circ\end{aligned}$$

Therefore, the value of the power factor angle θ_2 is 31.79° .

Step 8 of 17

Calculate the apparent power absorbed by load 2.

$$S_2 = \frac{P_2}{\cos \theta_2}$$

Substitute 16 kW for P_2 and 31.79° for θ_2 .

$$\begin{aligned}S_2 &= \frac{16 \text{ k}}{0.85} \\ &= 18.824 \text{ kVA}\end{aligned}$$

Therefore, the value of the apparent power S_2 is, 18.824 kVA.

Step 9 of 17

Calculate the reactive power absorbed by the load 2.

$$Q_2 = S_2 \sin \theta_2$$

Substitute 18.824 kVA for S_2 and 31.79° for θ_2 .

$$\begin{aligned}Q_2 &= (18.824 \text{ k}) \sin(31.79^\circ) \\ &= (18.824 \text{ k})(0.5268) \\ &= 9.917 \text{ kVAR}\end{aligned}$$

Therefore, the value of the reactive power Q_2 is 9.917 kVAR.

Step 10 of 17

Calculate the complex power absorbed by the load 2.

$$S_2 = P_2 + jQ_2$$

Substitute 9.917 kVAR for Q_2 and 16 kW for P_2 .

$$S_2 = (16 \text{ k} + j9.917 \text{ k}) \text{ VA}$$

Therefore, the value of the complex power absorbed by the load 2 is,

$$(16 \text{ k} - j9.917 \text{ k}) \text{ VA}.$$

Step 11 of 17

Calculate the current passing through the load 2.

$$I_2 = \left(\frac{S_2}{V} \right)^*$$

Substitute $(16 \text{ k} + j9.917 \text{ k}) \text{ VA}$ for S_2 and $440 \angle 0^\circ \text{ V}$ for V .

$$\begin{aligned} I_2 &= \left(\frac{16 \text{ k} + j9.917 \text{ k}}{440 \angle 0^\circ} \right)^* \\ &= (36.364 + j22.539)^* \\ &= 36.364 - j22.539 \\ &= 42.783 \angle -31.79^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current flowing through load 2 is, $42.783 \angle -31.79^\circ \text{ A}$.

Step 12 of 17

Calculations for load-3:

Calculate the power factor angle at load 3.

$$\begin{aligned} \theta_3 &= \cos^{-1}(0.866) \\ &= 30^\circ \end{aligned}$$

Therefore, the value of the power factor angle θ_3 is 30° .

Step 13 of 17

Calculate the apparent power absorbed by load 3.

$$S_3 = \frac{Q_3}{\sin \theta_3}$$

Substitute 15 kVAR for Q_3 and 30° for θ_3 .

$$\begin{aligned} S_3 &= \frac{15 \text{ K}}{\sin 30^\circ} \\ &= \frac{15 \text{ k}}{0.5} \\ &= 30 \text{ kVA} \end{aligned}$$

Therefore, the value of the apparent power S_3 is, 30 kVA .

Step 14 of 17

Calculate the real power absorbed by the load 3.

$$P_3 = S_3 \cos \theta_3$$

Substitute 30 kVA for S_3 and 30° for θ_3 .

$$\begin{aligned} P_3 &= (30 \text{ k}) \cos(30^\circ) \\ &= (30 \text{ k})(0.866) \\ &= 25.98 \text{ kW} \end{aligned}$$

Therefore, the value of the real power P_3 is 25.98 kW .

Step 15 of 17

Calculate the complex power absorbed by the load 3.

$$S_3 = P_3 + jQ_3$$

Substitute 15 kVAR for Q_3 and 25.98 kW for P_3 .

$$S_3 = (25.98 \text{ k} + j15 \text{ k}) \text{ VA}$$

Therefore, the value of the complex power absorbed by the load 3 is,

$$(25.98 \text{ k} + j15 \text{ k}) \text{ VA}.$$

Step 16 of 17

Calculate the current passing through the load 3.

$$I_3 = \left(\frac{S_3}{V} \right)^*$$

Substitute $(16 \text{ k} + j9.917 \text{ k}) \text{ VA}$ for S_2 and $440\angle 0^\circ \text{ V}$ for V .

$$\begin{aligned} I_3 &= \left(\frac{25.98 \text{ k} + j15 \text{ k}}{440\angle 0^\circ} \right)^* \\ &= (59.046 + j34.091)^* \\ &= 59.046 - j34.091 \\ &= 68.181\angle -30^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current flowing through load 3 is, $68.181\angle -30^\circ \text{ A}$.

Step 17 of 17

Calculate the value of the current I_o .

$$I_o = I_1 + I_2 + I_3$$

Substitute $68.181\angle -30^\circ \text{ A}$ for I_3 , $42.783\angle -31.79^\circ \text{ A}$ for I_2 and $31.492\angle 30^\circ \text{ A}$ for I_1 .

$$\begin{aligned} I_o &= 31.492\angle 30^\circ + 42.783\angle -31.79^\circ + 68.181\angle -30^\circ \\ &= 129.32\angle -18.43^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_o is, $129.32\angle -18.43^\circ \text{ A}$.